Anupam Dore

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which concerns the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

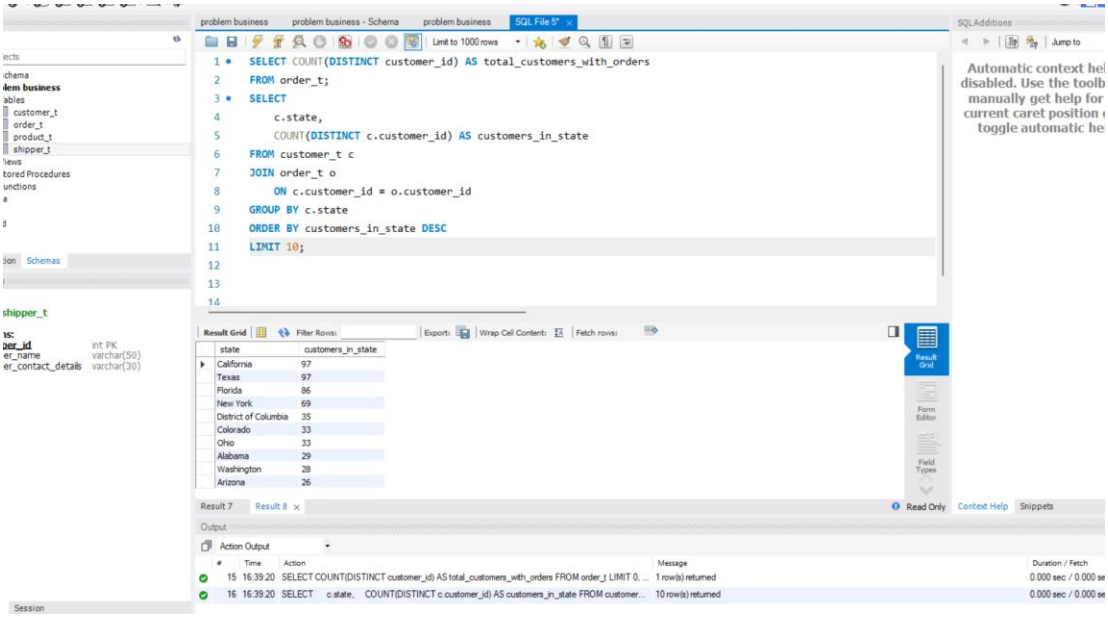
As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

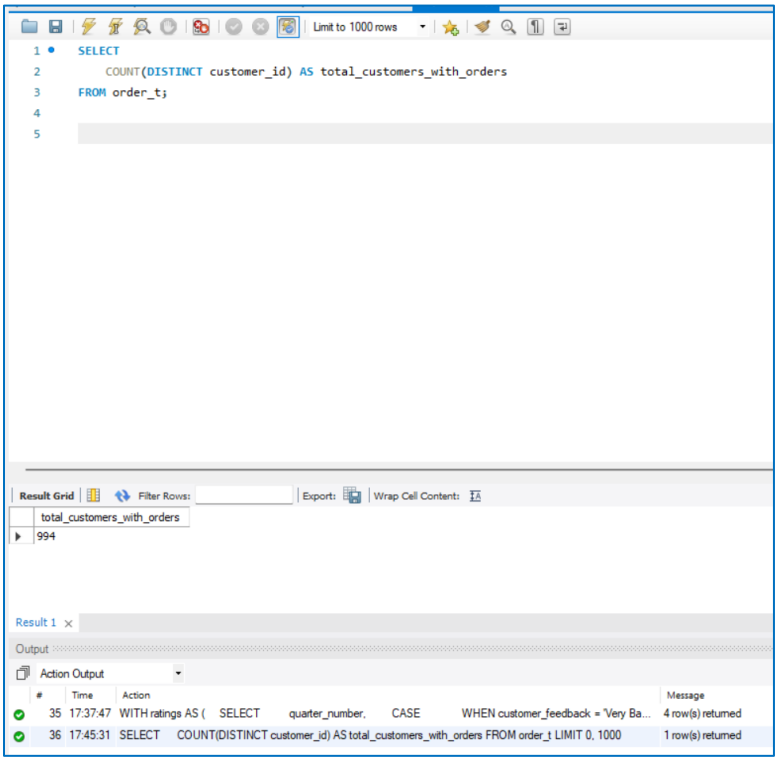
Question 1-- Find the total number of customers who have placed orders. What is the distribution of customers across states?

OUTPUT:

```
SELECT COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM order_t;
SELECT
   c.state,
   COUNT(DISTINCT c.customer_id) AS customers_in_state
FROM customer_t c
JOIN order_t o
   ON c.customer_id = o.customer_id
GROUP BY c.state
```

ORDER BY customers_in_state DESC LIMIT 10;





OUTPUT:

SELECT
COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM order_t;

Observations and Insights:

California (97) and Texas (97) are the largest customer bases, showing equal and exceptionally strong demand.

- These two states represent a significant portion of total customers, making them critical for revenue stability and growth.

- High customer volume suggests strong brand presence and potential for premium product upselling.
- District of Columbia (35) stands out for its small geographic size yet high customer count, suggesting a high-value urban demographic.
- Colorado (33) and Ohio (33) show balanced, stable demand, ideal for regional marketing and logistics optimization.

Alabama (29), Washington (28), and Arizona (26) represent developing markets with room for growth.

- These states may respond well to:
- Localized promotions
- Faster shipping options
- Loyalty or referral programs
- Improving customer experience in these regions could significantly increase order volume.

Question 2: -- Which are the top 5 vehicle makers preferred by the customers?

OUTPUT:

```
SELECT
   p.vehicle_maker,
   COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN product_t p
   ON o.product_id = p.product_id
JOIN customer_t c
   ON o.customer_id = c.customer_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

problem business | problem business - Schema | problem business | SQL File 6*

Limit to 1000 rows

```sql
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN product_t p
    ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

Result 1

Output

Action Output

#	Time	Action	Message
16	16:39:20	SELECT c.state, COUNT(DISTINCT c.customer_id) AS customers_in_state FROM customer...	10 row(s) returned
17	16:53:34	SELECT p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count FROM ord...	5 row(s) returned

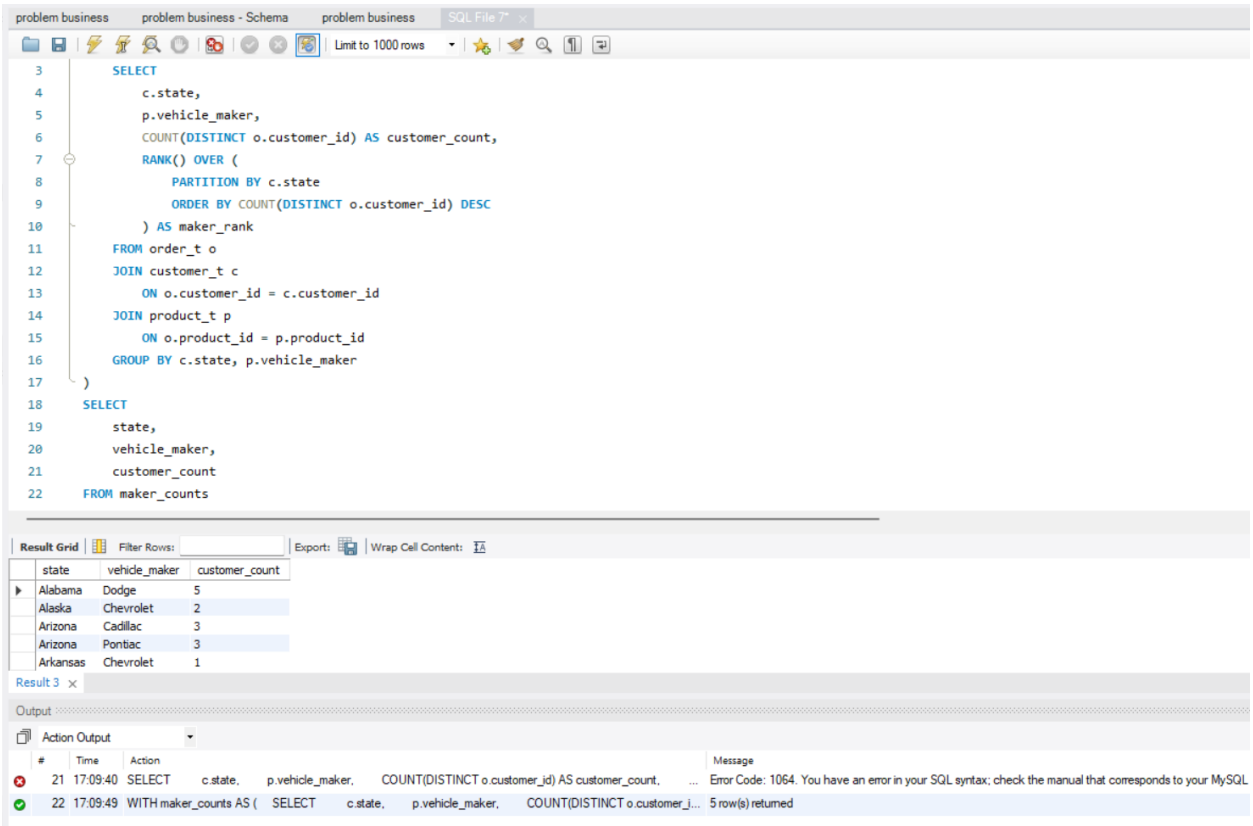
Observations and Insights:

- With 83 customers, Chevrolet clearly leads the market.
- This indicates strong brand loyalty and wide appeal across customer segments.
- The company should prioritize Chevrolet for inventory planning and promotional campaigns.
- 63 customers show Ford has a solid and reliable customer base.
- Although behind Chevrolet, Ford remains a major contributor to overall sales volume.
- Ford-focused marketing could help reduce the gap with Chevrolet.
- With 52 customers, Toyota maintains a competitive presence.
- This suggests customers value Toyota's reputation for reliability and efficiency.
- Toyota is a good candidate for targeted offers or seasonal promotions.
- Both Dodge and Pontiac have 50 customers, showing nearly identical demand.
- This indicates a balanced mid-tier segment where customers are open to multiple alternatives.
- Small incentives (discounts, financing options) could shift preference between these two brands.

Question 3: -- Which is the most preferred vehicle maker in each state?

OUTPUT:

WITH maker_counts AS (SELECT c.state, p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count, RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT o.customer_id) DESC) AS maker_rank FROM order_t o JOIN customer_t c ON o.customer_id = c.customer_id JOIN product_t p ON o.product_id = p.product_id GROUP BY c.state, p.vehicle_maker) SELECT state, vehicle_maker, customer_count FROM maker_counts WHERE maker_rank = 1 ORDER BY state LIMIT 5;



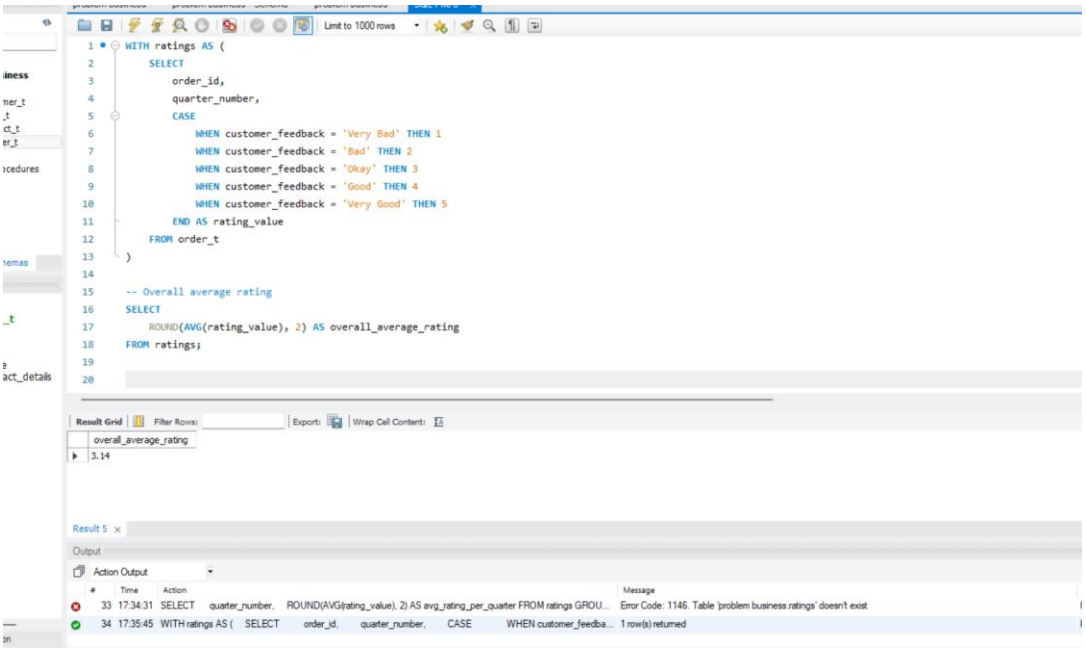
Observations and Insights:

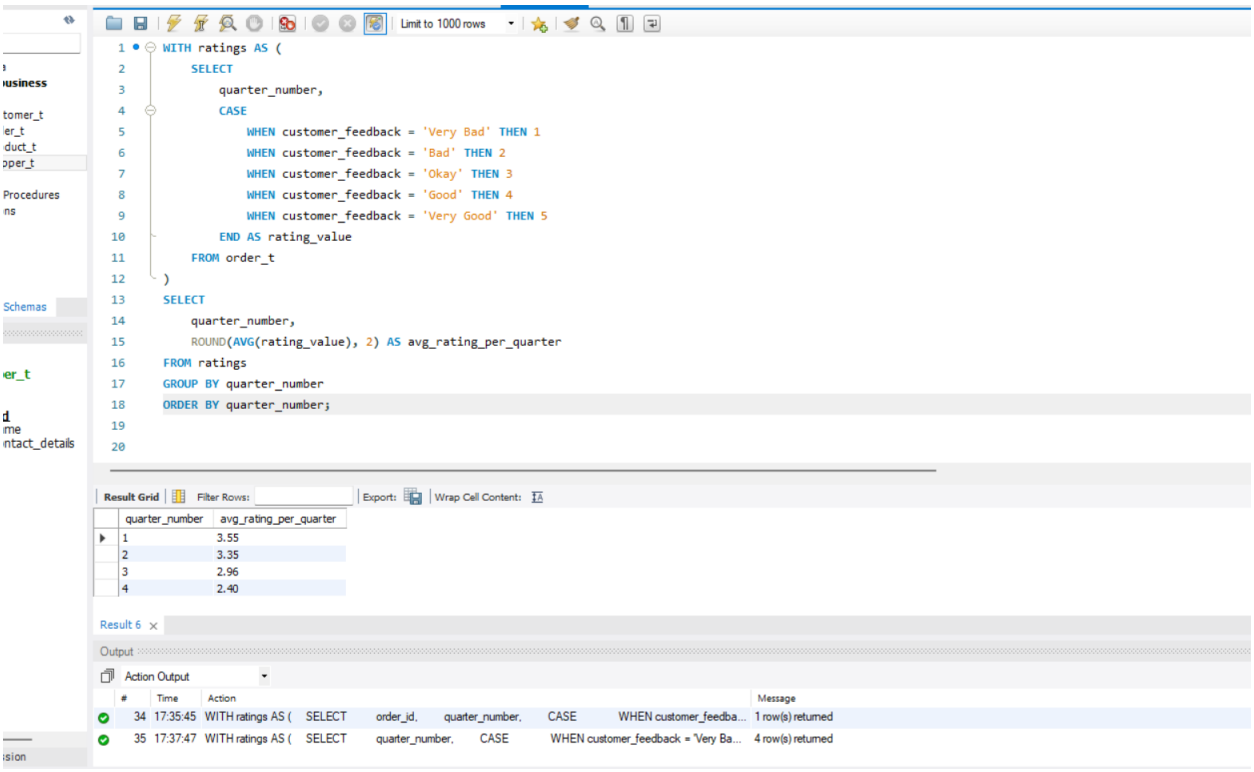
- Some states have very low customer counts, so the “top brand” is based on small numbers.
- Chevrolet appears multiple times, showing it has broad appeal across different states.
- States like Arizona show more variety in customer choices, while states like Alaska and Arkansas have limited data.
- The results highlight how customer preferences can vary a lot depending on the state and the number of active customers.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

OUTPUT:

```
WITH ratings AS (
   SELECT
      quarter_number,
      CASE
         WHEN customer_feedback = 'Very Bad' THEN 1
         WHEN customer_feedback = 'Bad' THEN 2
         WHEN customer_feedback = 'Okay' THEN 3
         WHEN customer_feedback = 'Good' THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating_value
   FROM order_t
)
SELECT
   quarter_number,
   ROUND(AVG(rating_value), 2) AS avg_rating_per_quarter
FROM ratings
GROUP BY quarter_number
ORDER BY quarter_number;
```





Observations and Insights:

- Customer satisfaction starts strong in Q1 and Q2 but declines sharply in Q3 and Q4.
- The downward trend suggests that the business should focus on identifying and fixing issues that emerged in the second half of the year.
- Improving consistency across quarters could help raise the overall rating above the current 3.14 average.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more

dissatisfied over time?

OUTPUT:

SELECT quarter_number,

```
-- Very Bad %
ROUND(
    (SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 2
) AS very_bad_pct,
```

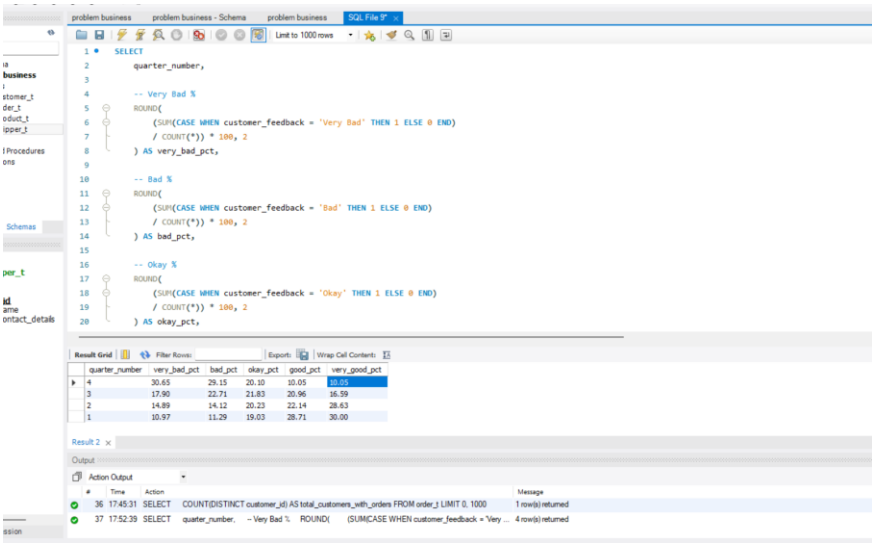
```
-- Bad %
ROUND(
    (SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 2
) AS bad_pct,

-- Okay %
ROUND(
    (SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 2
) AS okay_pct,

-- Good %
ROUND(
    (SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 2
) AS good_pct,

-- Very Good %
ROUND(
    (SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END)
    / COUNT(*)) * 100, 2
) AS very_good_pct
```

FROM order_t GROUP BY quarter_number ORDER BY quarter_number;



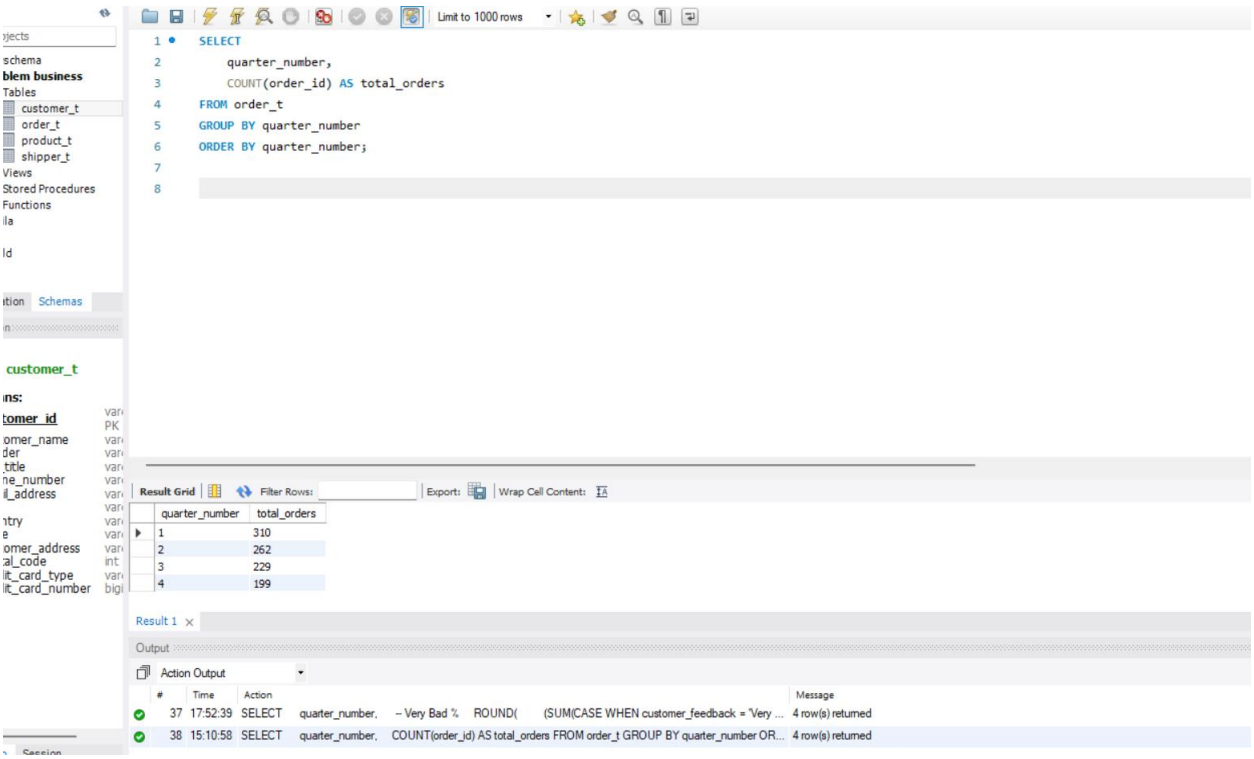
Observations and Insights:

- In Quarter 1, positive feedback (“Good” + “Very Good”) was 58.71%, while negative feedback (“Very Bad” + “Bad”) was only 22.26%.
  Customers were mostly satisfied at the beginning of the year.
- In Quarter 2, positive feedback dropped slightly to 50.77%, and negative feedback rose to 29.01%. Early signs of declining satisfaction started to appear.
- Quarter 3, negative feedback jumped to 40.61%, and “Very Good” dropped to 16.59%.
  This quarter shows a clear shift toward dissatisfaction.
- Quarter 4, negative feedback peaked at 59.80%, while positive feedback fell drastically to 20.10%.
  This was the worst-performing quarter, with the highest dissatisfaction levels.

Question 6: -- What is the trend of the number of orders by quarter?

OUTPUT:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```



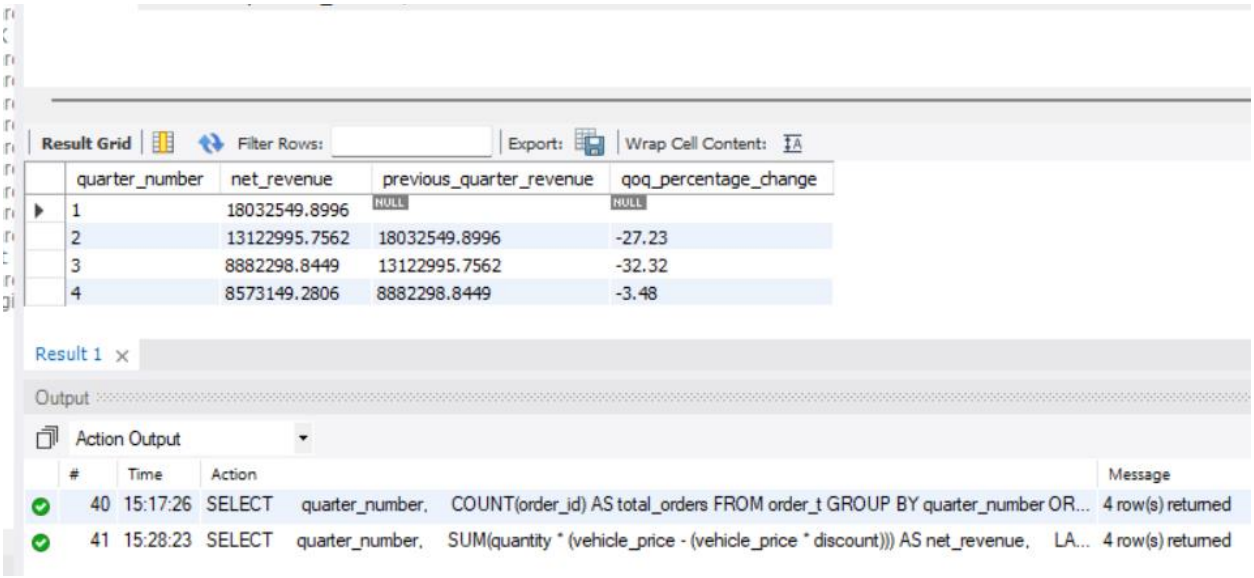
Observations and Insights:

- Quarter 1 had the highest number of orders (310), suggesting strong customer interest at the beginning of the year.
- Quarter 2 dropped to 262 orders, a noticeable decrease that may reflect mid-year saturation or reduced promotions.
- Quarter 3 continued the downward trend with 229 orders, possibly due to seasonal factors or shifting customer priorities.
- Quarter 4 had the lowest order count (199), which could indicate end-of-year fatigue, budget constraints, or reduced marketing efforts.
- Overall, the pattern suggests that early-year quarters perform better, and the business may benefit from targeted campaigns in Q3 and Q4 to reverse the decline.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

OUTPUT:

```
SELECT quarter_number, SUM(quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue,
LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number) AS
previous_quarter_revenue, ROUND( ( (SUM(quantity * (vehicle_price - (vehicle_price * discount))) -
LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number) ) /
LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number) ) *
100, 2) AS qoq_percentage_change FROM order_t GROUP BY quarter_number ORDER BY
quarter_number;
```



Observations and Insights:

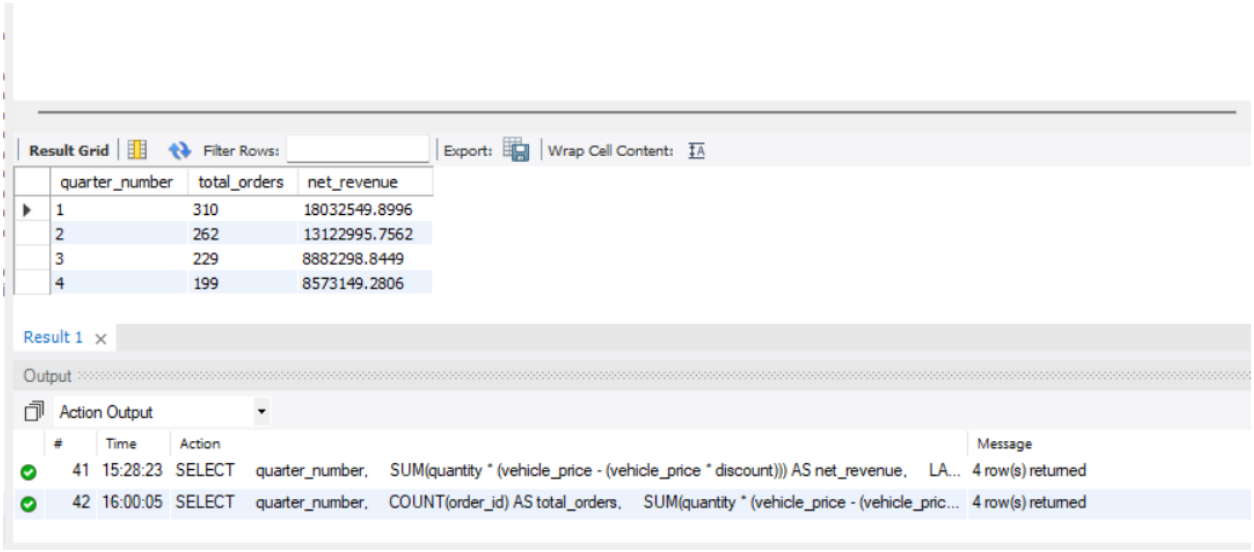
- Quarter 1 generated the highest net revenue of $18.03M, driven by strong customer demand and high order volume.

- Quarter 2 dropped to $13.12M, a 27.23% decline, signaling a slowdown in sales momentum.
- Quarter 3 fell further to $8.88M, a 32.32% drop, indicating deeper challenges in customer engagement or product appeal.
- Quarter 4 stabilized slightly at $8.57M, with only a 3.48% decline, suggesting the downward trend may be flattening
- The company should invest in targeted marketing and promotions in Q2 and Q3 to counter mid-year revenue drops.
- Consider analyzing product-level performance to identify which vehicles or models are losing traction.
- Explore customer feedback trends to understand satisfaction issues that may impact repeat purchases.
- Since Q4 shows signs of stabilization, the business could leverage loyalty programs or bundled offers to boost year-end sales.
- Overall, the company needs a data-driven strategy to reverse revenue decline and sustain growth across all quarters.

Question 8: What is the trend of net revenue and orders by quarters?

OUTPUT:

SELECT quarter_number, COUNT(order_id) AS total_orders, SUM(quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue FROM order_t GROUP BY quarter_number ORDER BY quarter_number;



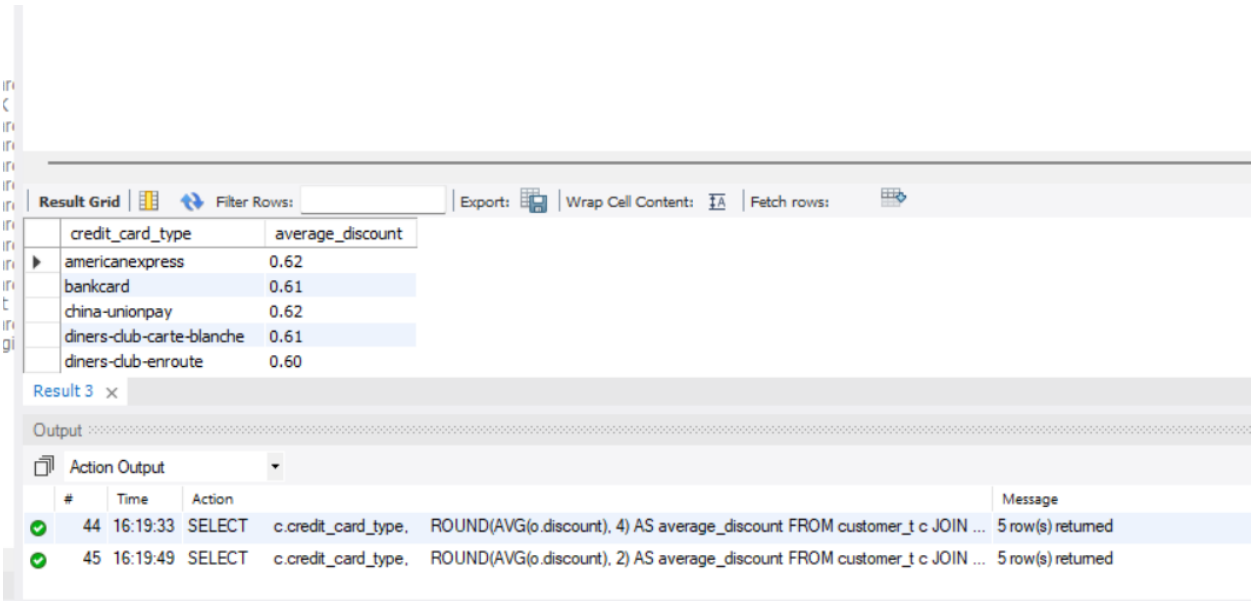
OBSERVATIONS AND INSIGHTS:

- Strong Q1 performance reflects high customer engagement and successful early-year sales strategy.

- Q2 and Q3 show steep declines in both orders and revenue, suggesting mid-year drop-offs in demand or operational efficiency.
- Q4 stabilizes revenue, but order volume continues to fall — indicating fewer transactions but possibly higher-value purchases.
- The parallel decline in both metrics signals a need to reassess customer retention, product appeal, and seasonal strategy.
- Revive mid-year performance with targeted campaigns, loyalty programs, and bundled offers to boost Q2–Q3 engagement.
- Use customer feedback and product-level data to identify friction points and optimize the buying experience.
- Explore shipper and ship mode data to improve delivery efficiency and reduce churn.
- Consider launching limited-time promotions in Q3 and Q4 to counter seasonal fatigue and stimulate urgency.

Question 9: What is the average discount offered for different types of credit cards?

OUTPUT:
```
SELECT
   c.credit_card_type,
   ROUND(AVG(o.discount), 2) AS average_discount
FROM customer_t c
JOIN order_t o
   ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY c.credit_card_type
LIMIT 5;
```



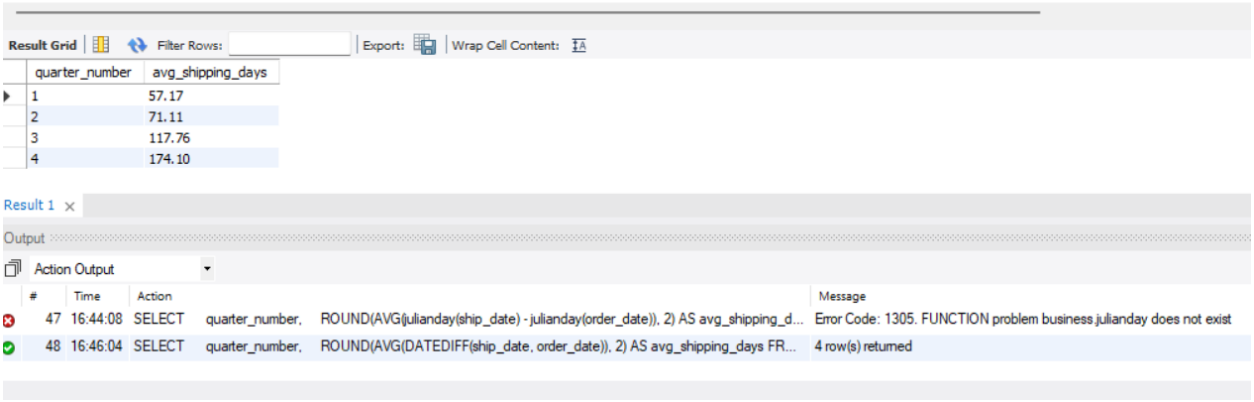
Observations and Insights:

- The average discount across all credit card types is consistent, ranging between 0.60% and 0.62%.
- American Express and China UnionPay customers receive the highest average discounts, possibly reflecting premium customer targeting or strategic partnerships.
- Diners Club Enroute shows the lowest average discount, which may indicate lower transaction volumes or less promotional focus.
- Consider segmenting customers by credit card type to identify high-value groups and tailor promotions accordingly.
- Explore whether higher discounts correlate with higher order volumes or revenue — this could justify deeper incentives for certain cardholders.
- Use this insight to optimize discount strategies and align them with customer behavior, loyalty, and profitability.
- If premium cardholders are driving more revenue, the company could expand targeted offers or loyalty perks to strengthen retention

Question 10: What is the average time taken to ship the placed orders for each quarter?

OUTPUT:

```
SELECT
   quarter_number,
   ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_shipping_days
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
```



Business Metrics Overview

TOTAL REVENUE: **$124,714,086.32**
TOTAL ORDERS: **1000**
TOTAL CUSTOMERS: **994**
AVERAGE RATING: **3.13**
LAST QUARTER REVENUE: **23,346,779.63**
LAST QUARTER ORDERS: **199**
AVERAGE DAYS TO SHIP: **97.96**
% GOOD FEEDBACK: **21.50**

BUSINESS RECOMENDATIONS:

1. With an average shipping time of nearly 98 days, customers are waiting far too long to receive their vehicles. This directly connects to the low overall rating (3.13) and the very low percentage of good feedback (21.5%). Faster fulfillment through better shipper partnerships, inventory planning, or regional distribution would immediately improve satisfaction and repeat purchase rates.
2. Strengthen Customer Experience Programs to Reverse the Declining Ratings Trend. Quarterly ratings drop from 3.55 → 2.40, showing a clear downward trend in customer sentiment. Introducing structured follow-ups, clearer communication during shipping, and proactive issue resolution can rebuild trust. Even a small improvement in ratings can significantly increase retention, especially since you have 994 unique customers but only 1,000 orders, meaning most customers buy only once.
3. Capitalize on Q4 Momentum by Expanding High-Demand Products. Last quarter generated $23.3M in revenue — almost 19% of total annual revenue — showing strong seasonal demand. Doubling down on the vehicle models, colors, or price

ranges that performed best in Q4 can help replicate that success earlier in the year. This could include targeted promotions, optimized inventory, or bundling strategies.

This SQL project helped me:

- Translate raw data into strategic business insights
- Use aggregation and window functions to track performance trends
- Think like a business analyst — not just coding — by aligning SQL logic with real-world decisions
- Communicate clearly with stakeholders through structured, data-driven reporting